Insurance Platform KPI Framework | Refined

- **01 Business outcomes and scale**
  - Primary KPI: issued policy count B01
  - Primary KPI: issued-policy premium B02
  - Coverage scale: total and policy/policyholder averages B03/B06/B07
  - Premium structure: policy/policyholder averages B04/B05
  - Submission and receipts: accepted applications B09 / collected premium B08
- **02 Application funnel and efficiency**
  - Quote request → valid quote → application transition → accepted submission → issued policy
  - Key drivers: step/end-to-end conversion F01/F02
  - Non-conversion and duration F03/F04
  - Mode comparison: manual/automatic quotation and entry paths F05/F06
  - See “03 Business flows” for detailed paths
- **03 User scale and retention**
  - New registrations U01
  - Active DAU/WAU/MAU U02
  - Exact-day D1/D7/D30 retention U03
  - Stickiness: average DAU/MAU U04
- **04 Feature usage**
  - Key-button clicks/users P01/P02
  - Feature adoption and button CTR P03/P04
  - Share actions/users P05
  - Lookup/search: page-level adoption and results P03/P06
- **05 User profiles and behavioral preferences**
  - Platform users: role, channel, organization, region, device
  - Policyholders: person/organization, region, purchase frequency, premium band
  - Insured parties: person/organization, count or covered-object type; separate from policyholders
  - Preferences: pages/paths, lookup/search, issuance mode, sharing
  - Product features: insurance line, product, coverage, sum insured, premium, term
- **06 Experience and service quality**
  - Failure counts/rate/causes E01/E02/E03
  - Contexts: application, online service, identity verification, registration/login
  - Online service completion and duration E04/E05
  - Guardrails: workflow failure rate E02 and service P90 duration E05
  - Consolidate application conversion and paths under the funnel
- **07 Claims analysis (optional)**
  - Closed and paid claims C01
  - Claim payments made C02
  - Insurance-line mix of closed paid claims C03
  - Provisionally move original undefined transaction scale into claims analysis; confirm intended meaning
- **08 Segmentation and data governance**
  - Shared dimensions: time, platform, channel, product, role, mode, organization, region
  - Frontend events measure actions; backend states confirm outcomes
  - Align IDs, deduplication, currency, timezone, observation windows, freshness
  - See “02” for definitions and “04” for conventions and terminology

Metric definitions and calculations

- Business outcomes and scale
  - B01 Issued policy count
    - Calculation / definition：Distinct policy_id values with backend-confirmed issuance during the reporting period
    - Measurement scope：Use first successful issuance time; application submission is not issuance
    - Purpose：Primary KPI: measures completed business volume
    - Proposed data source：Policy system: issuance status/time, premium, sum insured, policyholder
    - Refined from source metrics
  - B02 Total premium on issued policies
    - Calculation / definition：Sum of premiums confirmed at first issuance for B01 policies
    - Measurement scope：Group by currency; distinct from collected, earned, or net premium; report later cancellations and endorsements separately
    - Purpose：Primary KPI: measures monetary scale; review alongside B01 and E02
    - Proposed data source：Policy system: issuance status/time, premium, sum insured, policyholder
    - Refined from source metrics
  - B03 Total sum insured on issued policies (comparable products)
    - Calculation / definition：Sum insured across the subset of B01 policies with comparable coverage definitions
    - Measurement scope：Fix product scope, currency, and base/rider aggregation rules; do not simply add heterogeneous coverage limits
    - Purpose：Tracks coverage scale; avoid a single ranking across heterogeneous insurance lines
    - Proposed data source：Policy system: issuance status/time, premium, sum insured, policyholder
    - Refined from source metrics
  - B04 Average premium per policy
    - Calculation / definition：B02 / B01
    - Measurement scope：Same issuance cohort, currency, and product scope; show N/A when denominator is zero
    - Purpose：Tracks policy value and product-mix shifts
    - Proposed data source：Policy system: issuance status/time, premium, sum insured, policyholder
    - Refined from source metrics
  - B05 Average premium per policyholder
    - Calculation / definition：B02 / distinct policyholder_id values associated with B01
    - Measurement scope：Policyholders may be people or organizations; denominator is not platform users or insured persons
    - Purpose：Clarifies the population behind the original per-person premium metric
    - Proposed data source：Policy system: issuance status/time, premium, sum insured, policyholder
    - Refined from source metrics
  - B06 Average sum insured per policy (comparable products)
    - Calculation / definition：B03 / distinct policy count within the B03 subset
    - Measurement scope：Denominator includes only the comparable policies used in B03
    - Purpose：Tracks coverage per policy
    - Proposed data source：Policy system: issuance status/time, premium, sum insured, policyholder
    - Refined from source metrics
  - B07 Average sum insured per policyholder (comparable products)
    - Calculation / definition：B03 / distinct policyholder_id values within the B03 subset
    - Measurement scope：Per-person is defined here as per policyholder; per-insured coverage for group policies requires a separate metric and insured-level data
    - Purpose：Prevents mixing platform users, policyholders, and insured persons
    - Proposed data source：Policy system: issuance status/time, premium, sum insured, policyholder
    - Refined from source metrics
  - B08 Collected premium (proposed addition)
    - Calculation / definition：Sum of confirmed premium receipts during the period, deduplicated by payment transaction
    - Measurement scope：Use receipt time and currency; report refunds separately and do not mix with B02
    - Purpose：Reconciles payments with policy issuance
    - Proposed data source：Payment system: confirmed receipts, currency, refunds
    - Proposed addition
  - B09 Successfully submitted application count
    - Calculation / definition：Distinct application_id values accepted by the backend during the period
    - Measurement scope：Use server acceptance events; original application/H5 submit clicks represent intent only
    - Purpose：Preserves submission-stage volume while separating submission from issuance
    - Proposed data source：Application system: application ID and backend acceptance status
    - Refined from source metrics
- Application funnel and efficiency
  - F01 Step-to-step conversion rate
    - Calculation / definition：Cases reaching B after A within the same mature cohort / cases reaching A
    - Measurement scope：Deduplicate by linked business case ID; fix observation window W; compute user conversion separately by user_id
    - Purpose：Diagnoses bottlenecks between quote request, quote delivery, application transition, submission, and issuance
    - Proposed data source：Event logs, business status tables, and quote/application/policy links
    - Refined from source metrics
  - F02 Quote-request-to-issuance conversion rate
    - Calculation / definition：Quote requests producing at least one issued policy within W / quote requests in the same mature initiation cohort
    - Measurement scope：Include initiated requests without quotes or with later failures; count a request with multiple policies as one converted request
    - Purpose：Key driver: measures end-to-end completion
    - Proposed data source：Event logs, business status tables, and quote/application/policy links
    - Refined from source metrics
  - F03 Step non-conversion rate
    - Calculation / definition：1 – F01 for the same A→B transition
    - Measurement scope：Use mature cohorts only; distinguish exit, rejection, timeout, and pending cases; non-conversion is not always abandonment
    - Purpose：Locates unfinished stages and their causes
    - Proposed data source：Event logs, business status tables, and quote/application/policy links
    - Proposed addition
  - F04 Process duration P50/P90
    - Calculation / definition：Median and 90th percentile of duration between specified start and completion events
    - Measurement scope：Separate request-to-quote, submission-to-issuance, and request-to-issuance; state elapsed versus working time; show pending age separately
    - Purpose：Identifies slow workflows; review with unfinished-case counts to avoid completed-case bias
    - Proposed data source：Event logs, business status tables, and quote/application/policy links
    - Refined from source metrics
  - F05 Manual/automatic quotation success rate (proposed addition)
    - Calculation / definition：Distinct requests receiving a valid quote within W / distinct requests submitted for quotation
    - Measurement scope：Segment by final quotation mode; separately report manual takeover and automatic fallback; use mature processing cohorts
    - Purpose：Compares quotation modes
    - Proposed data source：Quotation system: requests, results, modes, fallback history
    - Proposed addition
  - F06 Entry usage by issuance mode
    - Calculation / definition：Count entry clicks, distinct users, and distinct business cases by issuance mode
    - Measurement scope：Display the three grains separately; do not also classify copy, batch, temporary-cover conversion, or renewal as standard new business
    - Purpose：Preserves original mode-level click and user counts with business-case context
    - Proposed data source：Product events: clicks, impressions, shares, qualifying actions
    - Refined from source metrics
- User scale and retention
  - U01 New registered users
    - Calculation / definition：Distinct user_id values first successfully registered during the period
    - Measurement scope：Exclude test, employee, and bot accounts; report new visitors separately
    - Purpose：Measures new-user volume
    - Proposed data source：Identity system: first successful registration time and account type
    - Refined from source metrics
  - U02 Active users: DAU/WAU/MAU
    - Calculation / definition：Distinct user_id values performing at least one defined qualifying business action per day/week/month
    - Measurement scope：Suggested actions: lookup, search, quote request, submission, and service actions; exclude passive impressions; WAU uses calendar weeks
    - Purpose：Measures meaningful usage; version the qualifying-action list
    - Proposed data source：Product events: clicks, impressions, shares, qualifying actions
    - Refined from source metrics
  - U03 D1/D7/D30 retention
    - Calculation / definition：Users in a registration-date cohort active on day N / users in that registration cohort
    - Measurement scope：N=1/7/30; exact-day retention; include only fully observed cohorts; distinct from policy renewal
    - Purpose：Tracks return usage and ongoing value
    - Proposed data source：Identity system and qualifying-action logs
    - Refined from source metrics
  - U04 User stickiness
    - Calculation / definition：Average daily DAU during the month / MAU for that month
    - Measurement scope：Use the same action list, timezone, and user scope; do not substitute one day's DAU for the month
    - Purpose：Measures usage frequency; interpret by user role in infrequent insurance workflows
    - Proposed data source：Product events: clicks, impressions, shares, qualifying actions
    - Refined from source metrics
- Feature usage
  - P01 Key-button click count
    - Calculation / definition：Valid click events on defined key buttons
    - Measurement scope：Deduplicate by event_id; clicks are not page views; group by page_id/button_id
    - Purpose：Clarifies the original button-click PV label
    - Proposed data source：Product events: clicks, impressions, shares, qualifying actions
    - Refined from source metrics
  - P02 Key-button clicking users
    - Calculation / definition：Distinct user_id values clicking the specified button
    - Measurement scope：Use a separately labeled visitor_id measure for anonymous traffic; do not simply combine with logged-in users
    - Purpose：Preserves click UV with an explicit identity grain
    - Proposed data source：Product events: clicks, impressions, shares, qualifying actions
    - Refined from source metrics
  - P03 Key-feature adoption rate
    - Calculation / definition：Active users using the feature / active users eligible and able to use it in the same period
    - Measurement scope：Fix page_id/feature_id and eligibility scope; calculate lookup and search separately
    - Purpose：Compares feature reach across pages and user groups
    - Proposed data source：Product events: clicks, impressions, shares, qualifying actions
    - Refined from source metrics
  - P04 Key-button click-through rate (CTR)
    - Calculation / definition：Button impression instances with at least one click / valid button impression instances
    - Measurement scope：Link clicks to impressions by impression_id; count visible impressions only; do not estimate CTR without exposure events
    - Purpose：Separates CTR from adoption and avoids distortion from repeated clicks
    - Proposed data source：Product events: clicks, impressions, shares, qualifying actions
    - Refined from source metrics
  - P05 Share actions and sharing users
    - Calculation / definition：Share-event count and distinct sharing user_id values
    - Measurement scope：Use confirmed shares where receipts exist; otherwise label share initiated and do not infer recipient delivery
    - Purpose：Measures sharing behavior
    - Proposed data source：Product events: clicks, impressions, shares, qualifying actions
    - Refined from source metrics
  - P06 Search result availability rate (proposed addition)
    - Calculation / definition：Valid search requests returning at least one valid result / all valid search requests
    - Measurement scope：Deduplicate by search_request_id; keep timeouts and API failures in the denominator and report causes separately
    - Purpose：Adds search quality context beyond clicks
    - Proposed data source：Search logs: request ID, result count, errors
    - Proposed addition
- Experience and service quality
  - E01 Workflow failure count
    - Calculation / definition：Distinct failed request_id values among valid business requests
    - Measurement scope：Retries use new request_id values; separately report affected business cases; segment application, service, verification, and login
    - Purpose：Preserves the original failure-count metric
    - Proposed data source：Backend request logs: request ID, terminal outcome, error code, completion time
    - Refined from source metrics
  - E02 Workflow failure rate (proposed addition)
    - Calculation / definition：E01 / valid requests with a terminal outcome in the same period
    - Measurement scope：Use outcome-completion time; terminal outcomes include success, failure, and confirmed timeout; show pending separately; match E01 scope
    - Purpose：Quality guardrail: interpret alongside request volume
    - Proposed data source：Backend request logs: request ID, terminal outcome, error code, completion time
    - Proposed addition
  - E03 Failure-cause distribution
    - Calculation / definition：Failed requests assigned to a primary cause / E01
    - Measurement scope：Assign one primary error_code per request; retain unknown causes; classify validation, business rejection, payment, permission, system, and network causes
    - Purpose：Diagnoses application, service, identity-verification, and registration/login issues
    - Proposed data source：Backend request logs: request ID, terminal outcome, error code, completion time
    - Refined from source metrics
  - E04 Online service completion rate
    - Calculation / definition：Service requests completed successfully within W / requests in the same mature initiation cohort
    - Measurement scope：Deduplicate by service_request_id; define funnels separately for each service type
    - Purpose：Preserves the original online-service funnel
    - Proposed data source：Online service system: request ID, initiation/completion status and time
    - Refined from source metrics
  - E05 Online service duration P50/P90
    - Calculation / definition：Duration percentiles from service initiation to successful completion
    - Measurement scope：Compute on completed requests; separately show unfinished counts and age; compare like service types
    - Purpose：Quality guardrail: monitors slow service cases
    - Proposed data source：Online service system: request ID, initiation/completion status and time
    - Refined from source metrics
- Claims analysis (optional)
  - C01 Closed and paid claim count (optional)
    - Calculation / definition：Distinct claim_id values closed during the period with cumulative paid amount greater than zero
    - Measurement scope：Use closure time; show zero-payment, denied, and reopened claims separately; requires claims-system data
    - Purpose：Clarifies the operational meaning of completed claims
    - Proposed data source：Claims system: claim status, payment transactions, insurance line
    - Proposed addition
  - C02 Claim payments made (optional)
    - Calculation / definition：Sum of successful claim payment transactions during the period, deduplicated by payment transaction
    - Measurement scope：Use payment time and currency; report recoveries and reversals separately; distinct from cumulative payments on C01 claims
    - Purpose：Provisionally interprets the original undefined amount total as claim payments; confirm intended meaning
    - Proposed data source：Claims system: claim status, payment transactions, insurance line
    - Refined from source metrics
  - C03 Insurance-line mix of closed paid claims (optional)
    - Calculation / definition：C01 claim count by insurance line / total C01 count within the same scope
    - Measurement scope：Assign each multi-line claim to one primary line; retain unknown lines; this is not a loss ratio
    - Purpose：Preserves the original completed-claim product characteristics
    - Proposed data source：Claims system: claim status, payment transactions, insurance line
    - Refined from source metrics

Business flows and common funnel

- **Common funnel (proposed state mapping)**
  - Quote requested → valid quote delivered → application-transition page entered → application accepted → policy issued
  - Separate page entry from backend success; buttons labeled quote/issuance or agreement issuance are entry labels
  - Track payment branches separately; non-payment-first does not imply waived premium or active coverage
  - Allow mode-specific skipped steps; compute conversion only across applicable stages
- **Issuance modes and entries (source paths preserved)**
  - R01 Standard new business: direct start
    - Start quote request → enter quote information
    - Preserves standard-product inquiry; merges the empty new-business node into the route category
  - R02 Favorite-product quote shortcut
    - Favorite-product quote button → enter quote information
    - Shares downstream flow with R01; record the entry source separately
  - R03 Copied quote request
    - Click copy → enter quote information → downstream quotation/application
    - Link the new quote ID to the source ID; downstream mapping is proposed
  - R04 “免核” product issuance (internal category)
    - Click quote → dedicated quote-entry page → downstream processing
    - Keep a separate page_id; provisionally interpret as manual-underwriting exemption; confirm automatic underwriting versus full exemption
  - R05 Agreement-based issuance: issuance-button entry
    - Search agreement number/name or policy number → click issuance → quote discussion
    - Track separately from R06; confirm the business meaning of agreement
  - R06 Agreement-based issuance: record-number entry
    - Search agreement number/name or policy number → click record number → agreement-based issuance
    - Preserve the original route; do not assume the record number is a policy number
  - R07 Batch application transition: application-button entry
    - Search quote-request number/policyholder/insured → click apply → application-transition page
    - Record batch_id and child-case IDs; measure batch and child-case success separately
  - R08 Batch application transition: quote-number entry
    - Click quote-request number → quote discussion
    - Separate from R07; track downstream steps per child case
  - R09 Temporary-cover conversion: product-lookup entry
    - Lookup product/coverage start date → product details → convert → quote entry
    - Link temporary-cover and final policy IDs; avoid double counting in B01
  - R10 Temporary-cover conversion: temporary-record entry
    - Search temporary-cover number → convert → quote entry
    - Same business mode as R09; retain distinct entry sources
  - R11 Renewal entry
    - Renewal tab → search policy number/insured/policyholder → select product → optional “中斷险”/channel details → start quote request → quote entry
    - Retain the internal label “中斷险” pending clarification; link original policy and renewal application; do not equate user retention with renewal
- **713 platform flows (source label retained)**
  - W01 Original 713 WEB flow
    - Request quote → enter/edit quote information → quote discussion → underwriting and quotation
    - Source ends at quotation; transition, submission, and issuance are proposed additions requiring backend-state mapping
  - H01 713 H5: insurance-line path
    - Set/change channel → select insurance line → quote/issuance entry → enter quote information → request quote → manual/automatic quotation → application transition → submit application → payment-first/other application success
    - Add issuance confirmation after the original application-success state; configure payment requirements by payment_mode
  - H02 713 H5: agreement path
    - Set/change channel → select agreement → agreement-based issuance → quote discussion → manual/automatic quotation → application transition → submit application → payment-first/other application success
    - Remove the stray final s in the source; record application and issuance success separately
- **Service and exception paths**
  - Online service: initiate → input/validate → submit → accept → complete (proposed template)
  - Exceptions: validation failure, quotation rejection, payment failure, timeout, exit, cancellation, retry
  - When one request produces multiple applications/policies, deduplicate conversion by request and volume by each relevant ID